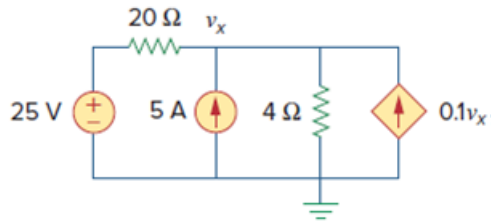


Problem

Use superposition to find v_x in the circuit of Fig. 4.11.

Figure. 4.11:



Step-by-step solution

Step 1 of 4

Refer to Figure 4.11 in the textbook.

There are two independent sources, each of them contribute to the dependent source, let

$$v_x = v_{x1} + v_{x2}$$

Where v_{x1} and v_{x2} are the contributions due to 25 V voltage source and 5 A current source respectively.

Apply superposition theorem to the circuit.

Set the 5 A current source to zero by replacing it with an open circuit, to obtain v_{x1} as shown in Figure 1.

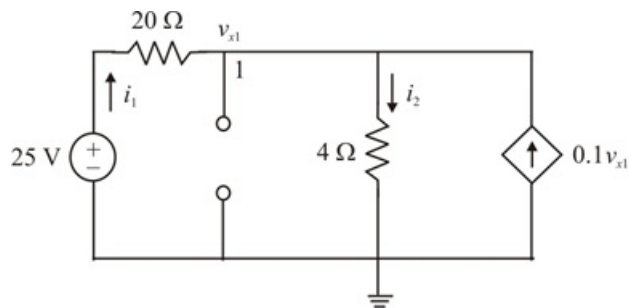


Figure 1

Step 2 of 4

Apply Kirchhoff's current law at node with node voltage v_{x1} of circuit in Figure 1.

$$i_1 + 0.1v_{x1} = i_2$$

$$\frac{25 \text{ V} - v_{x1}}{20 \Omega} + 0.1v_{x1} = \frac{v_{x1}}{4 \Omega}$$

$$25 \text{ V} - v_{x1} + 2v_{x1} = 5v_{x1}$$

$$5v_{x1} + v_{x1} - 2v_{x1} = 25 \text{ V}$$

$$4v_{x1} = 25 \text{ V}$$

$$v_{x1} = 6.25 \text{ V}$$

Step 3 of 4

Set the 25 V voltage source to zero by replacing it with a short circuit, to obtain v_{x2} as shown in Figure 2.

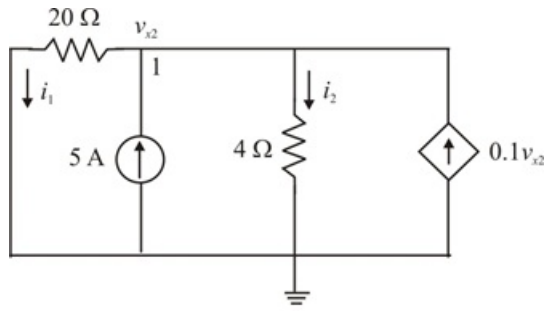


Figure 2

Step 4 of 4

Apply Kirchhoff's current law at node with node voltage v_{x2} of circuit in Figure 2.

$$i_1 + i_2 = 0.1v_{x2} + 5 \text{ A}$$

$$\frac{v_{x2}}{20 \Omega} + \frac{v_{x2}}{4 \Omega} = 0.1v_{x2} + 5 \text{ A}$$

$$v_{x2} + 5v_{x2} = 2v_{x2} + 100 \text{ V}$$

$$v_{x2} + 5v_{x2} - 2v_{x2} = 100 \text{ V}$$

$$4v_{x2} = 100 \text{ V}$$

$$v_{x2} = 25 \text{ V}$$

Calculate the node voltage v_x .

$$v_x = v_{x1} + v_{x2}$$

$$= 6.25 \text{ V} + 25 \text{ V}$$

$$= 31.25 \text{ V}$$

Therefore, the voltage v_x is 31.25 V.